

§4.4.

P3. Solution.

$$(a) A^T A = \begin{bmatrix} 16 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 16 \end{bmatrix}$$

$$(b) A^T A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

P6. Solution:

$$Q = Q_1 Q_2.$$

$$\begin{aligned} Q^T Q &= (Q_1 Q_2)^T (Q_1 Q_2) \\ &= Q_2^T (Q_1^T Q_1) Q_2 \\ &= I \end{aligned}$$

P8. $(q_1^T b) q_1 + (q_2^T b) q_2$

P11. Solution:

$$(a) \tilde{q}_1 = a = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 5 \end{bmatrix}, \quad \|\tilde{q}_1\|^2 = 100$$

$$\tilde{q}_2 = b - \left(\frac{\tilde{q}_1^T b}{\tilde{q}_1^T \tilde{q}_1} \right) \tilde{q}_1 = \begin{bmatrix} -6 \\ 6 \\ 8 \\ 8 \end{bmatrix} - \left(\frac{100}{100} \right) \begin{bmatrix} 1 \\ 3 \\ 4 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} -7 \\ 3 \\ 4 \\ -5 \end{bmatrix}, \quad \|\tilde{q}_2\|^2 = 100$$

$$\Rightarrow q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|} = \begin{bmatrix} 1/10 \\ 3/10 \\ 2/5 \\ 1/2 \end{bmatrix}, \quad q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|} = \begin{bmatrix} -7/10 \\ 3/10 \\ 2/5 \\ -1/2 \end{bmatrix}$$

$$(b) q_1^T b = \frac{1}{10}, \quad q_2^T b = -\frac{7}{10}$$

$$\begin{aligned} p &= (q_1^T b) q_1 + (q_2^T b) q_2 \\ &= \begin{bmatrix} 1/2 \\ -9/50 \\ -6/25 \\ 2/5 \\ 0 \end{bmatrix} \end{aligned}$$

P16. Solution:

$$\hat{x} = \frac{a^T b}{a^T a} = \frac{14}{7^2} = 2/7$$

$$\tilde{q}_1 = a = \begin{bmatrix} 4 \\ 5 \\ 2 \end{bmatrix}, \quad \|\tilde{q}_1\|^2 = 49.$$

$$\tilde{q}_2 = b - \left(\frac{\tilde{q}_1^T b}{\tilde{q}_1^T \tilde{q}_1} \right) \tilde{q}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - \left(\frac{2}{7} \right) \begin{bmatrix} 4 \\ 5 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} -1/7 \\ 4/7 \\ -4/7 \end{bmatrix}, \quad \|\tilde{q}_2\|^2 = 1.$$

$$\Rightarrow q_1 = \begin{bmatrix} 4/7 \\ 5/7 \\ 2/7 \end{bmatrix}, \quad q_2 = \begin{bmatrix} -1/7 \\ 4/7 \\ -4/7 \end{bmatrix}$$

P21. Solution:

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 0 \\ 1 & 3 \end{bmatrix}$$

$$\tilde{q}_1 = a_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \|\tilde{q}_1\|^2 = 4.$$

$$\tilde{q}_2 = a_2 - \left(\frac{\tilde{q}_1^T a_2}{\tilde{q}_1^T \tilde{q}_1} \right) \tilde{q}_1 = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - \left(\frac{2}{4} \right) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -5/2 \\ -1/2 \\ 5/2 \end{bmatrix}$$

$$\Rightarrow q_1 = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \quad q_2 = \begin{bmatrix} -5/2\sqrt{3} \\ -1/2\sqrt{3} \\ 5/2\sqrt{3} \end{bmatrix}, \quad \|\tilde{q}_2\|^2 = 13$$

$$\begin{aligned} p &= (q_1^T b) q_1 + (q_2^T b) q_2 \\ &= (-2) \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} + \left(\frac{26}{2\sqrt{3}} \right) \begin{bmatrix} -5/2\sqrt{3} \\ -1/2\sqrt{3} \\ 5/2\sqrt{3} \end{bmatrix} \\ &= \begin{bmatrix} -7/2 \\ -3/2 \\ -1/2 \\ 3/2 \end{bmatrix} \end{aligned}$$

P22. Solution:

$$A = a = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$B = b - \left(\frac{A^T b}{A^T A} \right) A = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - \left(\frac{0}{6} \right) \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$C = c - \left(\frac{A^T c}{A^T A} \right) A - \left(\frac{B^T c}{B^T B} \right) B$$

$$= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \left(\frac{9}{6} \right) \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \left(\frac{1}{2} \right) \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

S5.1.

P1. Solution:

$$\det(2A) = 2^4 \det A = 16 \cdot \left(\frac{1}{2}\right) = 8$$

$$\det(-A) = (-1)^4 \det A = \frac{1}{2}$$

$$\det(A^2) = (\det A) \cdot (\det A) = \frac{1}{4}$$

P9. Solution:

$$\det A = 1 \quad (\text{two row exchanges})$$

$$\det B = -\det \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = -\det \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$= -\det \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix} = -(-2) = 2$$

$$\det C = \det \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

P14. Solution:

$$i) \det \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 6 & 6 & 1 \\ -1 & 0 & 0 & 3 \\ 0 & 2 & 0 & 7 \end{bmatrix} = \det \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 2 & 3 & 3 \\ 0 & 2 & 0 & 7 \end{bmatrix}$$

$$= \det \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 6 \end{bmatrix} = 1 \cdot 2 \cdot 3 \cdot 6 = 36$$

$$ii) \det \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} = \det \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

$$= \det \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & 0 \\ 0 & 0 & \frac{4}{3} & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} = \det \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & 0 \\ 0 & 0 & \frac{4}{3} & -1 \\ 0 & 0 & 0 & \frac{5}{4} \end{bmatrix}$$

$$= 5$$

P15. Solution:

$$i) \det \begin{bmatrix} 10 & 1 & 20 & 1 & 30 & 1 \\ 10 & 2 & 20 & 2 & 30 & 2 \\ 10 & 3 & 20 & 3 & 30 & 3 \end{bmatrix} = \det \begin{bmatrix} 10 & 1 & 20 & 1 & 30 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

$$= \det \begin{bmatrix} 10 & 1 & 20 & 1 & 30 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} = 0$$

$$ii) \det \begin{bmatrix} 1 & t & t^2 \\ t & 1 & t \\ t^2 & t & 1 \end{bmatrix} = \det \begin{bmatrix} 1 & t & t^2 \\ 0 & 1-t^2 & t-t^3 \\ 0 & t-t^3 & 1-t^4 \end{bmatrix}$$

$$= \det \begin{bmatrix} 1 & t & t^2 \\ 0 & 1-t^2 & t-t^3 \\ 0 & 0 & 1-t^2 \end{bmatrix}$$

$$= (1-t^2)^2$$

P23. Solution:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, \det A = 4 \cdot 3 - 1 \cdot 2 = 10$$

$$\det A = (\det A)^2 = 100$$

$$\det A^{-1} = \frac{1}{\det A} = \frac{1}{10}$$

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (4-\lambda)(3-\lambda) - 2$$

$$= \lambda^2 - 7\lambda + 10$$

$$= (\lambda - 2)(\lambda - 5)$$

$$\Rightarrow \text{when } \lambda = 2 \text{ or } 5, \det(A - \lambda I) = 0.$$

P27. Solution:

$$\det A = \det \begin{bmatrix} 0 & a & 0 \\ 0 & 0 & b \\ c & 0 & 0 \end{bmatrix} = -\det \begin{bmatrix} c & 0 & 0 \\ 0 & 0 & b \\ 0 & a & 0 \end{bmatrix}$$

$$= \det \begin{bmatrix} c & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{bmatrix} = abc$$

$$\det B = -abcd \quad (\text{three row exchanges})$$

$$\det C = \det \begin{bmatrix} a & a & a \\ a & b & b \\ a & b & c \end{bmatrix} = \det \begin{bmatrix} a & a & a \\ 0 & b-a & b-a \\ 0 & b-a & c-a \end{bmatrix}$$

$$= \det \begin{bmatrix} a & a & a \\ 0 & b-a & b-a \\ 0 & 0 & c-b \end{bmatrix}$$

$$= a(b-a)(c-b)$$